Modified GAN Model for Traffic Missing Data Imputation

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ABSTRACT

Nowadays, more and more traffic research relies on plentiful intelligent transportation data. However, due to hardware, software, or environmental reasons, traffic data may face various missing value problems. Previous studies have tried to tackle this problem using different models. However, as the missing rate increases, some models do not perform well. As we know, the GAN (generative adversarial network) model has achieved great success in the field of computer vision. This paper attempts to comprehensively impute the missing values at different missing rates by transforming the traffic data into image data. We use the modified GAN model to make the imputation. Studies have shown that the modified GAN model performs well compared with other models at different missing rates.

Keywords: Generative adversarial network; Missing value imputation; Missing rate; Computer vision.

1. INTRODUCTION

With the increasing number of urban traffic vehicles, urban roads are facing increasingly serious traffic congestion problems. The intelligent transportation system provides a good idea for solving urban traffic congestion (Sładkowski and Pamuła, 2016). Especially, traffic big data is the foundation of the intelligent transportation system and has a profound impact on traffic management and government decision-making. However, due to various reasons such as hardware, software, infrastructure, data transmission(Shang et al., 2018), etc., traffic data is facing severe data loss problems, which have an adverse impact on the analysis of data. Therefore, the missing value problem in traffic data deserves our attention, and many scholars proposed numerous imputation models.

We can divide missing value imputation techniques into three categories:

(1) the simple imputation method, such as using the average or median value to fill the blank, or using the interpolation method if the missing rate is not too high.

(2) The other is the matrix completion method. For this method, the original data is organized into a data matrix, and each row or column is regarded as a data point. The most popular ones under this group include probabilistic principal component analysis (PPCA)(Li et al., 2013), Bayesian principal component analysis (BPCA) (Asif et al., 2013), singular value decomposition (SVD) (Arciniegas-Alarcón et al., 2014)imputation method, etc.

(3) The third category is the currently popular machine learning or deep learning approaches. Generally, the missing value is regarded as the attribute value to be predicted, and it is predicted using the attribute that is not missing. Several widely used methods include decision tree (DT) (Rahman and Islam, 2011), artificial neural network (ANN) (Shukur and Lee, 2015), fuzzy-C means(Tang et al., 2015), etc.

Based on the above, there have been many studies on the field of missing value imputation of traffic data. However, note that traffic data is unusual compared with other types of survey data, considering its recurrence and periodicity pattern. This pattern can be reflected by daily or weekly recurrence on the same road segment. Moreover, some different road segments may also have similar traffic patterns. Simple imputation methods like linear interpolation do not fully make use of the characteristics of traffic data. It only gets satisfactory results when the missing rate is low. When the miss rate increases, the imputation error will rise significantly or even become intolerable. The matrix-filling method or machine learning method such as ANN usually treats one row or one column of data as a data point, while ignoring that the column or row of the data matrix is highly correlated. Therefore, it is necessary to treat the data matrix as a whole and focusing on the overall and local characteristics to achieve better imputation results.

In the field of computer vision, the generative adversarial network (GAN) (Goodfellow et al., 2014) model was used for image classification and achieved great success. It divides the model into the generator part and discriminator part. The generator part tries to generate fake images, and the discriminator part tries to discriminate between the real and fake ones. By competing with each other, the GAN model achieves high accuracy of the final result. Inspiringly, to make full use of the correlation of traffic data on different dimensions, we can treat the traffic data matrix as a whole image. Through the image processing, we can create similar generator and discriminator parts used to make an imputation, and finally achieve promising good imputation results.

In this paper, we explore a modified GAN model's performance on an open traffic dataset, and compare the results with some conventional models and analyze the experimental results in different traffic situations. We organize our paper as below: in Section 2, the methodology of imputation. In Section 2, we describe the modified GAN model and the reference models. In Section 3, we conduct a case study in Guangzhou, China, and make a comprehensive analysis with several other approaches. Section 4 then concludes this paper.

2. METHODOLOGY

2.1 Framework of GAIN model

The framework of the modified GAN model-"GAIN" model(Yoon et al., 2018) is illustrated in Figure 1. This model is mainly composed of two parts, namely the generator and discriminator. The generator is primarily applied to generate the filled matrix $\widehat{\boldsymbol{X}}$, and its performance is measured by mean squared error (MSE) between the non-missing part of $\overline{\boldsymbol{X}}$ and data matrix $\widetilde{\boldsymbol{X}}$. The discriminator part can produce the estimated mask matrix $\widehat{\boldsymbol{M}}$, and its performance is represented by the cross-entropy between the estimated mask matrix $\widehat{\boldsymbol{M}}$ and the real mask matrix $\boldsymbol{M}$. In particular, the model introduces the hint matrix $\boldsymbol{H}$ to improve the ability of the discriminator. The detailed process of the model is described below.

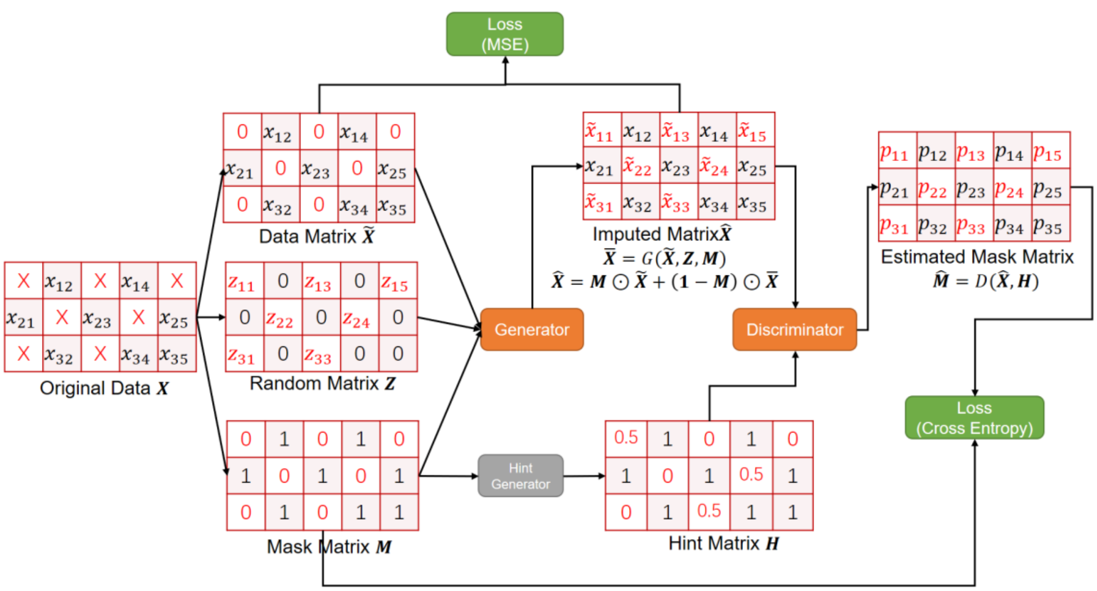


Figure 1. Framework of Generative Adversarial Imputation Network (GAIN)

Firstly, we construct the original data matrix $\boldsymbol{X}$ (with size $n \times p$), of which some entries are missing. We decompose $\boldsymbol{X}$ into three components: (1) The data matrix $\widetilde{\boldsymbol{X}}$ is constructed by simply replacing the missing entries of $\boldsymbol{X}$ with zeros; (2)

The random matrix $\boldsymbol{Z}$ are all zeros except replacing the missing entries with random numbers between 0 to 1; (3) The mask matrix $\boldsymbol{M}$ denotes whether the entries in $\boldsymbol{X}$ is missing. "0" denotes the corresponding entry in $\boldsymbol{X}$ is missing and "1" means the entry of X is not missing.

The generator G takes $\widetilde{\boldsymbol{X}}$, $\boldsymbol{Z}$ and $\boldsymbol{M}$ as input and output the initial imputation matrix $\overline{\boldsymbol{X}}$. Note that G output a value for each entry in $\overline{\boldsymbol{X}}$. We only need the imputed missing values. Therefore, we get the imputed matrix $\widehat{\boldsymbol{X}}$ by replacing each "×" in $\boldsymbol{X}$ with the corresponding value in $\overline{\boldsymbol{X}}$. The L_M is mean squared error (MSE) between the real data matrix $\widetilde{\boldsymbol{X}}$ and the corresponding non-missing part of $\overline{\boldsymbol{X}}$, which is $\boldsymbol{M} \odot \overline{\boldsymbol{X}}$. The mathematical form is:

$$\overline{\boldsymbol{X}} = G\left(\widetilde{\boldsymbol{X}}, \boldsymbol{M}, \boldsymbol{Z}\right) \tag{1}$$
$$\widehat{\boldsymbol{X}} = \boldsymbol{M} \odot \widetilde{\boldsymbol{X}} + (1 - \boldsymbol{M}) \odot \overline{\boldsymbol{X}} \tag{2}$$
$$L_M = MSE\left(\widetilde{\boldsymbol{X}}, \boldsymbol{M} \odot \overline{\boldsymbol{X}}\right) \tag{3}$$

where $\odot$ is the Hadamard product or element-wise multiplication. G is a function that ensures the out put $\overline{\boldsymbol{X}}$ has the same size as $\widetilde{\boldsymbol{X}}$. For example, a multi-layer perception or neural network could be a generator.

The hint generator takes the mask matrix $\boldsymbol{M}$ as input and randomly changes some zeros in missing entries to a fixed probability value that ranges from 0 to 1. Afterwards, the discriminator D takes $\widehat{\boldsymbol{X}}$ and $\boldsymbol{H}$ as inputs and output its judgement $\widehat{\boldsymbol{M}}$. The elements in $\widehat{\boldsymbol{M}}$ are from 0 to 1, and displays that the discriminator thinks whether the element in $\widehat{\boldsymbol{X}}$ is a real value or missing value. This judgement is evaluated by the cross entropy L_D between the estimated mask matrix $\widehat{\boldsymbol{M}}$ and original mask matrix $\boldsymbol{M}$. In mathematical form:

$$\widehat{\boldsymbol{M}} = D(\widehat{\boldsymbol{X}}, \boldsymbol{H}) \tag{4}$$
$$L_D = cross\ entropy(\boldsymbol{M}, \widehat{\boldsymbol{M}}) \tag{5}$$

Similarly, the discriminator is a function which output $\widehat{\boldsymbol{M}}$ with the same size as $\boldsymbol{M}$. It can be a neural network or some other dedicatedly designed functions. Moreover, the cross entropy (Shore and Johnson, 1980) of two matrices **A** and **B** is defined as:

$$cross\ entropy\ (\boldsymbol{A}, \boldsymbol{B}) = \sum_{ij} a_{ij} \log\left(b_{ij}\right) + \left(1 - a_{ij}\right) \log(1 - b_{ij}) \tag{6}$$

Based on the above all, we can iteratively update the parameters in the generator and discriminator, utilizing backpropagation skills. When the stop criterion is satisfied, we get our imputation result by the imputed result $\widehat{\boldsymbol{X}}$.

2.2 Comparisons models

We have also selected three imputation models to make a comparison with the GAIN model in our study. They are the multiple linear regression (MLR)(Leonard et al., 2005) method, the PPCA matrix completion(Li et al., 2013) method, the artificial neural network(ANN) (Shukur and Lee, 2015) approach. We briefly introduce their imputation process as below.

(1) MLR

Suppose the data point $\boldsymbol{x}$ can be divided into two parts: $\boldsymbol{x} = [\boldsymbol{x_{mis}}, \boldsymbol{x_{obs}}]$, where $\boldsymbol{x_{mis}}$ is the missing part and $\boldsymbol{x_{obs}}$ is the observed part. In this approach, the $\boldsymbol{x_{mis}}$ is supposed to be our prediction target. We establish the linear relationship between the missing attributes and non-missing attributes using the complete data records:

$$\boldsymbol{x_{mis}} = LR(\boldsymbol{x_{obs}}) \tag{7}$$

where LR is the multiple linear relationship. With this correlation, we can predict the missing part $\boldsymbol{x_{mis}}$ with the observed part $\boldsymbol{x_{obs}}$.

(2) PPCA

PPCA is a maximum-likelihood reformulation of PCA. Missing data can be imputed during this process. A brief description of PPCA in the probabilistic sense can be expressed as

$$\boldsymbol{x} = \boldsymbol{W}\boldsymbol{t} + \boldsymbol{\mu} + \boldsymbol{\epsilon} \tag{8}$$

where $\boldsymbol{x}$ is the data point with missing value, $\boldsymbol{t}$ is a latent vector of observed data, $\boldsymbol{W}$ is the projection matrix, $\boldsymbol{\mu}$ is the mean of $\boldsymbol{x}$, which keeps $\boldsymbol{x}$ with a zero mean. $\boldsymbol{\epsilon}$ is the white noise. If $\boldsymbol{x}$ is incomplete, $\boldsymbol{x}$ is divided into the missing part $\boldsymbol{x_{mis}}$ and $\boldsymbol{x_{obse}}$, and correspondingly, W can be written as $\boldsymbol{W} = [\boldsymbol{W_{mis}}, \boldsymbol{W_{obse}}]^T$. When $\boldsymbol{W_{mis}}$ is estimated, the missing part $\boldsymbol{x_{miss}}$ can be obtained by

$$\boldsymbol{x_{mis}} = \boldsymbol{W_{mis}}\widetilde{\boldsymbol{x}} \tag{9}$$

(3) ANN

Similarly, like MLR, the ANN approach also establishes the relationship between the missing part x_{mis} and the observed part x_{obs} using the complete data records. However, it uses the neural network to approximate the relationship rather than the linear regression. In mathematical form:

$$\boldsymbol{x_{mis}} = \boldsymbol{W_{mis}}\widetilde{\boldsymbol{x}} \tag{10}$$

where ANN is the neural network relationship. With this correlation, we can also predict the missing part $\boldsymbol{x_{mis}}$ with the observed part $\boldsymbol{x_{obs}}$.

3. CASE STUDY

3.1 Data set description

We use an open dataset from the OpenITS platform. This dataset includes traffic speed data of 214 road segments in Guangzhou, China. The period is two months, which ranges from August 1, 2016, to September 30, 2016. For each day in each segment, it includes 144 10-min time intervals. For example, the speed data of the 45th road segment on the 32nd day from 0:00 to 1:00 is displayed in Table 1.

Road_Id	Day_Id	Time_Id	Speed
45	32	1	53.121
45	32	2	52.989
45	32	3	54.141
45	32	4	53.919
45	32	5	51.648
45	32	6	53.237

Table 1. Sample of traffic speed data

For every road segment, we construct the weekly data matrix with size 7×144 to reflect the daily and weekly characteristics. In Figure 2, we exhibit three randomly selected weekly data matrix in the form of an image with 7×144 pixels. Thus, we have $214 \times 61/7$ images, which are applied to train and validate our model.

Apparently, Figure 2a and Figure 2b are less congested compared with Figure 2c. Additionally, for Figure 2a and b, there is no significant difference between the weekdays and the weekend. For comparison, Figure 2c obviously has morning and evening peaks on weekdays.

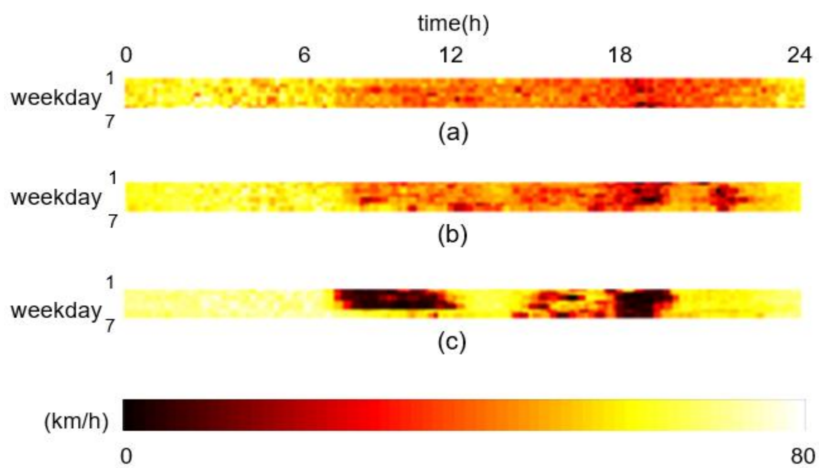


Figure 2. Sample image visualization of three data matrix

3.2 Experimental settings

Missing Patterns

Generally, there are three kinds of missing value patterns(Qu et al., 2008).

1) Missing Completely at Random (MCR): Whether the value is missing is entirely independent of other variables. It only depends on a fixed probability value, which is the missing rate.
2) Missing at Random (MR): The missing points are related to neighbor ones, usually in the form of a small group.
3) Missing at Determinate (MD): The missing points are in some particular pattern,which is generally caused by long-time malfunction of the equipment.

Generally, MD only happens when there is a longtime malfunction of detectors, which is not our topic here. In our research, for simplification, we deal with the MCR missing pattern. Under this missing pattern, we artificially set the missing rate from 0.1 to 0.6 to reflect the performance under different missing scenarios.

Evaluation Criteria

In our study, we use mean absolute error (MAE), root mean squared error (RMSE), and mean absolute percentage error (MAPE) to evaluate the average performance for the missing value imputation:

$$
\begin{aligned}
RMSE &= \sqrt{\frac{\sum_{m=1}^{M}\left(x_{org}^{m} - x_{est}^{m}\right)^2}{M}} \\
\mathrm{MAE} &= \frac{1}{M}\sum_{m=1}^{M}\left|\frac{x_{org}^{m} - x_{est}^{m}}{x_{org}^{m}}\right| \\
MAPE &= \frac{1}{M}\sum_{m=1}^{M}\left|x_{org}^{m} - x_{est}^{m}\right|
\end{aligned}
\tag{11}
$$

where M is the number of the missing points, x_{org}^{m} and x_{est}^{m} are the m-th original value and its estimation.

3.3 Model performance

For the exhibition, we assume that there is a 60% missing rate for each weekly data matrix, as shown in Figure 2. In Figure 3, at the 100th, 200th, 800th, and 1600th iteration, we randomly selected a data matrix for observation. At each iteration, we also randomly selected three imputation results and compared them with the real values. At the 100th iteration, the three imputation results are similar, but they are far away from the real data matrix. At the 200th and 800th iterations, the imputation results are closer to the real data matrix, which approves the effectiveness of the GAIN model. However, the values from 0:00 to around 6:00, from 22:00 to 24:00, are profoundly different from the real values. Nonetheless, in the 1600th iteration, the imputation results are highly similar to the real values, including the night time.

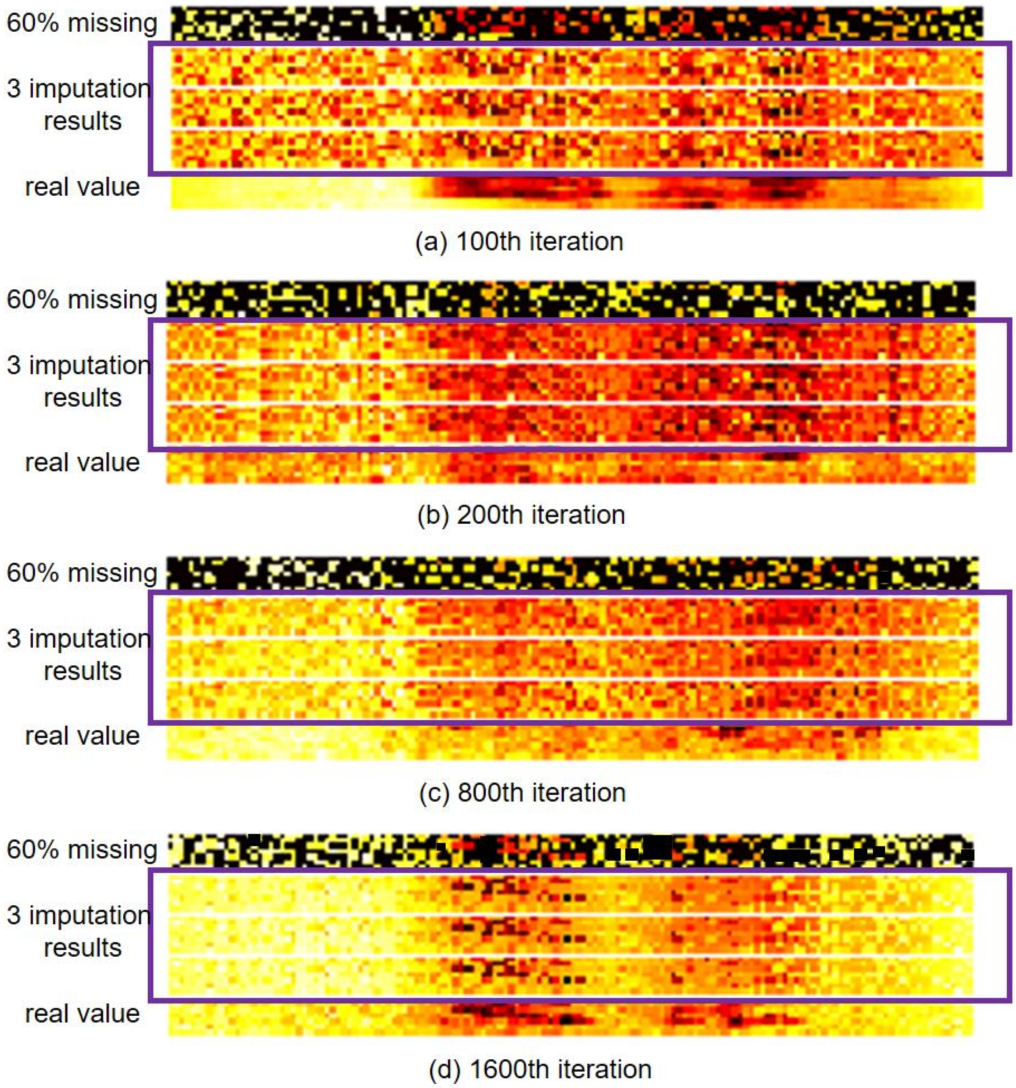


Figure 3. Observation of performance at 100, 200, 800, 1600th iterations

We further tested the imputation results under four different traffic patterns in Figure 4. We set the missing rate to be 0.5. Figure 4a and b depict the situations of one or two peaks. The imputation results can approximate the real values to a high degree. Figure 4c displayed the scenario that the road is free most of the time.

Figure 4d shows the condition when a road is always congested in the day. We find that the GAIN model can recognize the above four traffic patterns utilizing the non-missing values, and produce a good result after enough iterations. Even for hours late at night, the model could satisfactorily recover the original results too.

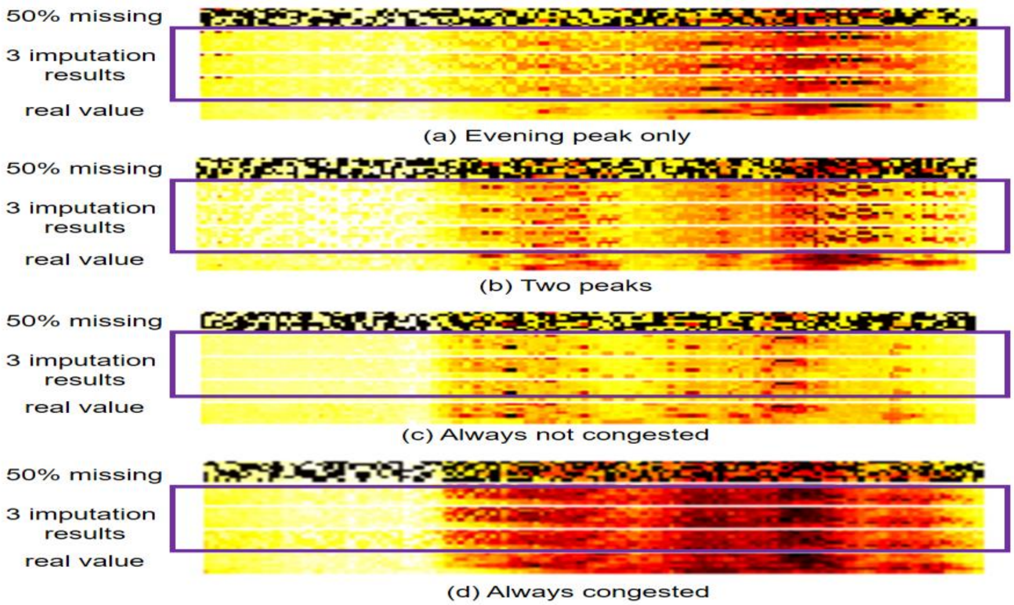
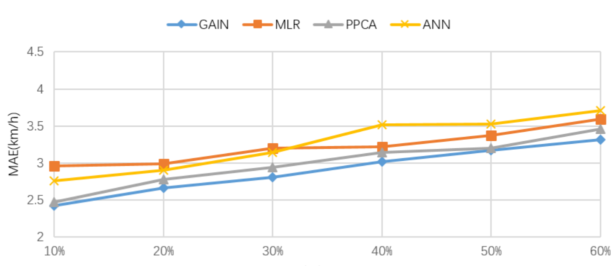


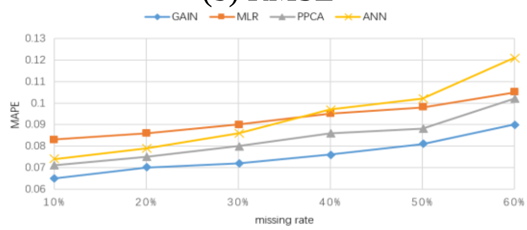
Figure 4. Example performance under four traffic patterns

To make an in-depth analysis, we have compared the model in our study with the other three models—MLR, PPCA, and ANN. The comparison result is shown in Figure 5. It can be seen that as the missing rate goes up, the imputation error of all four methods increases. However, the GAIN model performs better than the other three models under various missing rates. For example, when the missing rate increase from 10% to 60%, the MAPE only grows from 6.5% to 9%., which shows the model's robustness and imputation power. The experimental results of MLR or ANN are not good, because the relationship between the missing value and non-missing values may not be described by a simple linear relationship or the neural network.



(a) MAE

(b) RMSE



(c) MAPE

Figure 5. Performance comparison under different missing rates

4. CONCLUSION

With the rapid development of intelligent transportation systems, the analysis of traffic data is becoming more and more critical. However, plenty of traffic data faces more or less missing value problems, which harm the analysis and management of traffic data. This study applies the popular generative adversarial network (GAN) model and its variant – "GAIN" model in the field of computer vision to impute the missing values. The experimental result shows that the GAIN model can recover the original data through sufficient training steps. It also exhibits better high imputation ability under different traffic modes (one traffic peak, two traffic peaks, always not congested and always congested). For example, when the missing rate increases from 10% to 60%, MAPE only rises from 6.5% to 9%, thus showing good robustness and practicality.

Although the model performance is satisfactory in the task of traffic data imputation, the model requires a longer computation time. Meanwhile, this paper is only limited to the missing completely at random (MCR) situation. Therefore, future research should explore the application of the model under more missing patterns. Moreover, future research should also consider incorporating the spatial correlation to obtain better results.

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